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Relating soil organic matter composition to soil water repellency for soil biopore surfaces different in history from two Bt horizons of a Haplic Luvisol

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Abstract

The deposition of organic matter (OM), which is known for its high potential water repellency, on biopore walls can enhance preferential flow through these pores. In this study, OM composition determined with diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy was related to soil water repellency (SWR) determined with sessile drop method, Wilhelmy plate method, and sorptivity tests. We hypothesized that the chemical composition (in terms of potential wettability index; a) is related to the physical properties (i.e., contact angle) of biopore walls, (b) depends on the history of the biopore, and (c) differs from the bulk soil matrix. Thus, the main objective was to identify the relation between OM composition determined with DRIFT spectroscopy and SWR in structured soils. The experiments were carried out on biopores and their surrounding soil matrices, excavated from 2 depths of a haplic Luvisol, with 3 different biopore histories (i.e., root channels, earthworm burrows, and root channels that were short-term colonized by an earthworm). All measurements at intact biopore surfaces indicated a larger SWR at the surface of biopore walls as compared with the surrounding matrices and showed a higher proportion of hydrophobic functional groups. The OM composition determined with DRIFT spectroscopy correlated ($R^2 > .7$) with contact angles (sessile drop method) that is in line with results of both water sorptivity and Wilhelmy plate method for soils with reduced wettability. The surfaces of short-term colonized earthworm burrows had the most varying hydrophobic to hydrophilic components (A/B)-ratio of all investigated biopore surfaces depending on soil depth. For biopore surfaces at this depth, contact angles $>90^\circ$ were frequently observed. The results also indicate that earthworms can lower SWR by aggregate disruption.

KEYWORDS

biopores, DRIFT spectroscopy, earthworm burrow, organo-mineral coatings, root channel, undisturbed soil samples

1 | INTRODUCTION

Preferential flow paths such as earthworm burrows and root channels (Jarvis, 2007) play a major role for gas, water, and heat fluxes in soils. Beside these indirect influences of macropores on the fertility of structured soils, biopores may act as pathways for roots to propagate into deeper soil layers where water and nutrient uptake can be enhanced (Kautz et al., 2012). Organic matter (OM) is known for coating the outermost layer of mineral particles (Doerr, Shakesby, & Walsh, 2000; Ellerbrock & Gerke, 2004).

Elluvial processes, which (especially) occur in Luvisols (Ap/E/Bt/C horizon sequence), are characterized by leaching dissolved organic

substances and clay particles from the A and E horizon to the B horizon, respectively, resulting in clay(-organo-)coatings at the surfaces of soil aggregates. On the other hand, earthworms and roots exude numerous mostly organic compounds such as sugars, amino acids, and terpenes but also carbon dioxide (Jones, Nguyen, & Finlay, 2009; Scheu, 1991). The total amount and composition of the exudates varies with organism age, species, feeding, health status, and soil environmental conditions (Carvalhais et al., 2011; Christiansen-Weniger, Groneman, & van Veen, 1992; Jones et al., 2009). There is also scope to modify organic compounds released to biopore surfaces through plant breeding, as large differences between crop cultivars have been observed (Christiansen-Weniger et al., 1992). Furthermore, decayed

plant roots (debris and fine root hairs) increase the carbon content in the rhizosphere (for a review, see Six, Bossuyt, Degryze, & Denef, 2004). All these processes result in altered soil properties compared with the bulk soil (Nunan, Wu, Young, Crawford, & Ritz, 2003). In consequence, bioturbation by earthworms or roots can cause biopores with concentrated OM along their walls (Kautz et al., 2012). Both the outermost layer of soil aggregates and the surface of biopores are hotspots in terms of microbial activity (Hoang et al., 2016; Kuzyakov & Blagodatskaya, 2015) and interaction between percolating water, reactive solutes, and the soil matrix.

Aggregation next to biopores is influenced by two factors, both enhancing soil aggregation: (a) Organic compounds, sometimes coupled with the presence of decaying roots, may stick together mineral particles (Six, Frey, Thiet, & Batten, 2006) and (b) plants may enhance wetting and drying (Vetterlein & Marschner, 1993) and coincide shrinkage and swelling (Horn, Taubner, Wuttke, & Baumgartl, 1994; Peng & Horn, 2005). Beside other influences, aggregate development is also affected by the pH, because protons can act as a monovalent cation. Although the soil around roots can become more acidic (Uteau et al., 2015) due to the roots' release of CO_2 and H^+ , the soil around earthworm burrows can be more neutral/alkaline caused by the earthworms' release of CaCO_3 from specialized oesophageal glands that were discovered by Lankester (1865). Some earthworms are also able to affect the pH in their environment (Sizmur, Watts, Brown, Palumbo-Roe, & Hodson, 2011). However, bioturbation can also homogenize soil aggregates, for example, at high water contents (Horn et al., 1994), forming an unstructured mixture that release stabilized OM (Six et al., 2006) and an intermixture of aggregate surfaces with their interiors (Bossuyt, Six, & Hendrix, 2005). Such processes often result in heterogeneous distributions and compositions of OM at a small scale that in turn may influence soil properties such as contact angle (CA).

The wettability of intact biopore surfaces created by either earthworms or plant roots has been intensively studied (e.g., Leue, Eckhardt, Ellerbrock, Gerke, & Leinweber, 2016; Leue, Ellerbrock, & Gerke, 2010; Leue, Gerke, & Ellerbrock, 2013; Urbanek, Walsh, & Shakesby, 2014). It is often related to the soil organic carbon content (Urbanek et al., 2014; Zheng, Morris, Lehmann, & Rillig, 2016). But organic matter in soil (SOM) is a heterogeneous mixture of organic components containing hydrophilic (C=O) and hydrophobic (C-H) functional groups that can be determined with infrared spectroscopy (e.g., Ellerbrock & Gerke, 2004). The ratio between the C-H absorption band intensities (i.e., a measure for hydrophobic groups) and the C=O absorption band intensities (i.e., a measure for hydrophilic groups), the C-H/C=O-ratio, was found to be related to the wettability of soil samples (e.g., Ellerbrock, Gerke, Bachmann, & Goebel, 2005) and was defined as a potential wettability index (PWI; Leue et al., 2013). However, there is a lack in information about alteration in surface properties of both earthworm burrows and plant root channels induced by earthworm activity.

Beside the PWI, a couple of methods for the assessment of the wettability of soils are available (Bachmann, Woche, & Goebel, 2003). Although CAs can be measured directly with the sessile drop method, indirect measurements such as sorptivity tests or capillary rise method (CA derived by comparing, e.g., flow rates of ethanol with that of

water) or the time that droplet of an ethanol-water mixture of a certain molality need to penetrate (Bachmann et al., 2003; Moody & Schlossberg, 2010) represent the methods used in general. But there is a lack in relating spectral information on the PWI of SOM composition at intact surfaces with their wettability properties.

Our aim was to investigate the influence of earthworms, plant roots, and their effects on the PWI. Two experimental setups were used to measure the PWI on different scales. And we linked the PWI with a physical property: CAs determined from sessile drop method, sorptivity tests, and the Wilhelmy plate method.

We hypothesized that the chemical composition (in terms of PWI; a) is related to the physical properties (i.e., CA) of biopore walls, (b) depends on the history of the biopore, and (c) differs from the bulk soil matrix. Furthermore, we expected that the chemical composition of organic compartments is distributed more heterogeneously if measured on a smaller scale. We expected an increasing degree of scattering with increasing spatial resolution, reflecting the small-scaled heterogeneity of intact biopore walls or, more generally, of soils. With a view to soil ecology and, for example, hydraulic processes, a smaller-scaled analysis seemed useful, because biopore walls may contain large numbers of pinholes that can act as a preferential flow path from the biopore surface into the surrounding matrix. The quality of the OM of these pinholes can differ from those of biopore walls (Leue et al., 2016). Because these very small pinholes can have a huge hydraulic impact (described by the Hagen-Poiseuille law), a small-scaled analysis of the wettability of soils (or of related parameters, i.e., the PWI) is useful for characterizing hydraulic properties of pore walls. To the best of our knowledge, this is the first work that supplies results for PWI on two different scales (spatial resolution ranging from 0.018 to 0.79 mm^2).

2 | MATERIAL AND METHODS

2.1 | Soil material and sampling

In total 72 soil cores were excavated in September 2014 from a Haplic Luvisol (IUSS Working Group WRB, 2006) at the experimental area of Campus Klein-Altendorf (50°37'9" N 6°59'29" E, University of Bonn, Germany). The site is characterized by a maritime climate with

TABLE 1 Sand, silt, clay, and soil organic carbon (SOC) contents in g kg^{-1} soil, as well as, pH, and electrical conductivity (eC) in $\mu\text{S cm}^{-1}$ of the four pits where soil cores were excavated from a loess-derived Luvisol, Klein-Altendorf near Bonn, Germany

Pit	Depth	Sand	Silt	Clay	SOC	pH	eC
	m			g kg^{-1}		—	$\mu\text{S cm}^{-1}$
24	0.45–0.55	71.4	780	149	4.0	7.0	83
24	0.55–0.65	66.1	730	204	5.8	7.03	74
40	0.45–0.55	54.7	650	295	4.5	6.99	100
40	0.55–0.65	44.8	710	245	3.4	7.11	83
57	0.45–0.55	59.8	770	170	3.9	7.02	81
57	0.55–0.65	44.6	740	215	4.1	7.0	71
74	0.45–0.55	58.9	660	281	4.0	6.98	104
74	0.55–0.65	40.0	700	260	3.8	7.02	77

temperate humid conditions (9.6 °C mean annual temperature, 625 mm annual rainfall). Some main soil properties are listed in Table 1. The Bt horizon is characterized by accumulated clay, leached from the A horizon (0–0.27 m) and the E horizon (0.27–0.41 m).

The soil samples were excavated from four pits (see Table 1) of a totally randomized trial (Kautz et al., 2014). Chicory (*Cichorium intybus* L. "Puna," 5 kg ha⁻¹ seeding rate) had been grown in these trials for the last 3 years. With its herringbone or monopodial branching root systems, *C. intybus* L. is known for its ability to penetrate deeply into the subsoil exploring for water and nutrient supply (Kautz et al., 2012). After the roots decay, large (diameter ≥ 5 mm), continuous biopores remain that can potentially be colonized by earthworms. Soil samples from two layers within the Bt horizon (0.41–1.15 m) were investigated: Bt-1 from 0.45–0.55-m and Bt-2 from 0.55–0.65-m depth.

From both horizons, soil cores of 3 cm in diameter and 10 cm in height were excavated such that each core contained a macroscopic biopore (≥5 mm in diameter). The type of the biopore was classified endoscopically (Kautz et al., 2014) according to its "History" as either:

- Colonized and/or created by a plant root (*C. intybus* L., [R])
- Colonized and/or created by an earthworm (*Lumbricus terrestris*, [EW])
- Colonized and/or created by a plant root followed by short-term colonization of *L. terrestris* (REW).

The latter one had been colonized by earthworms for a period of approximately 6 months. Earthworms were fed with ryegrass residues placed at the soil surface in this period. Before sampling, the epi-earthworms were removed with the octet method (Thielemann, 1986).

Because *L. terrestris* was the only deep-burrowing earthworm present at the studied site, all earthworm burrows were formed by this species. The sample cores were first equilibrated to a defined matric potential of $\psi_m = -30$ kPa, equals field capacity, and scanned with a micro-X-ray computed tomograph (μ CT Nanotom[®] 180; GE Sensing & Inspection Technologies GmbH, Wunstorf, Germany). Afterwards, subsamples were saturated again and drained to defined matric potentials (i.e., $\psi_m = -1$ kPa; -3 kPa; -6 kPa—using sand beds—or -30 kPa—with the help of ceramic plates. See Haas, Holthausen, Mordhorst, Lipiec, & Horn, 2016, for more information) for sorptivity measurements described later. Subsamples were cut off vertically (Figure 1) using a scalpel for all measurements. Subsamples were stored at 4 °C prior to the initial equilibration and before diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy.

2.2 | Some soil properties

Disturbed, air-dried, and sieved soil (<2 mm) was used for standard soil parameters. pH values were determined in 25 ml 0.01 M CaCl₂ (1:2.5 m/v, Blume, Stahr, & Leinweber, 2010). Inorganic carbon was determined by Scheibler analysis (Blume et al., 2010). Soil organic carbon content was calculated by subtracting the inorganic carbon concentration from total carbon concentration determined via dry combustion at 1,200 °C (Coulomat 702, Ströhlein instruments, Kaarst,

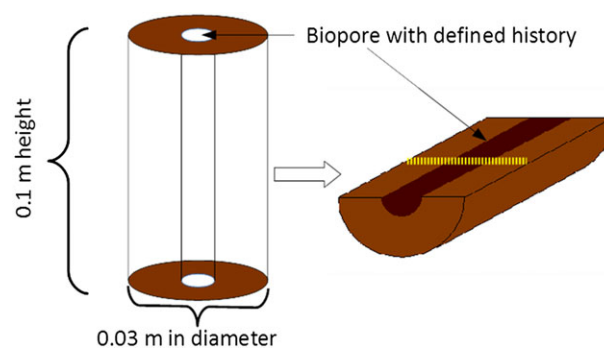


FIGURE 1 Sample preparation and exemplary transect (yellow, dotted line). Samples were cut off vertically, air-dried, and stored in a desiccator overnight, before diffuse reflectance infrared Fourier transform spectroscopy was performed along the transects. Results of measurements at the biopores' surface were pooled (=0-mm distance to biopore surface, Figures 3 and 4). The same was done for equidistant measurement points

Germany; Schlichting, Blume, & Stahr, 1995). Soil texture was determined according to Schlichting et al. (1995) by combined sieving and sedimentation processes (pipette method) after removal of cementing substances (HCl to remove CaCO₃; H₂O₂ for organic carbon oxidation; Na-pyrophosphate for dispersion).

2.3 | DRIFT spectroscopy

The preparation of the equilibrated subsamples for DRIFT spectroscopy required a thickness reduction of the subsamples by scraping off material from the external side of the soil column such that the samples thickness was about 0.01 m. The obtained thin intact soil samples were wrapped in tin foil (Ellerbrock, Gerke, & Böhm, 2009) and fixed on aluminium plates—with the external side located towards the aluminium plate. These thin intact soil samples were air-dried and stored in a desiccator over-dried silica gel for 16 hr, before DRIFT spectroscopy was performed. Measurements were performed at two scales:

(a) The DRIFT mapping technique (macro-DRIFT) performed with a Bio-Rad FTS 135 (Bio-Rad Corp., Hercules, CA, USA) combined with an XY-positioning table (Pike, Madison, USA, adapted by Resultec, Illerkirchberg, Germany). The DRIFT mapping was carried out along transects across the biopores in 1-mm steps (Figure 2; transect length is the whole sample width). Here, two transects of each of four samples were investigated per history and depth. The infrared (IR) beam was separately focused at each measurement point with respect to surface elevation (Leue et al., 2013) to avoid microtopography effects on the spectra (Leue, Ellerbrock, Bänninger, & Gerke, 2010). Each macro-DRIFT spectrum was recorded by 16 co-added scans.

(b) DRIFT microscopy (micro-DRIFT) used a Cary 660 combined with a Cary 610 microscope (both from Agilent Technologies, Santa Clara, CA, USA). Transects that cover the first 2.5 mm next to the biopore wall were measured in 0.15-mm steps (see scheme in Figure 2). Two transects of each of two samples were investigated per history and depth. For the micro-DRIFT measurements, no separate focussing of each measurement point was done to ensure that

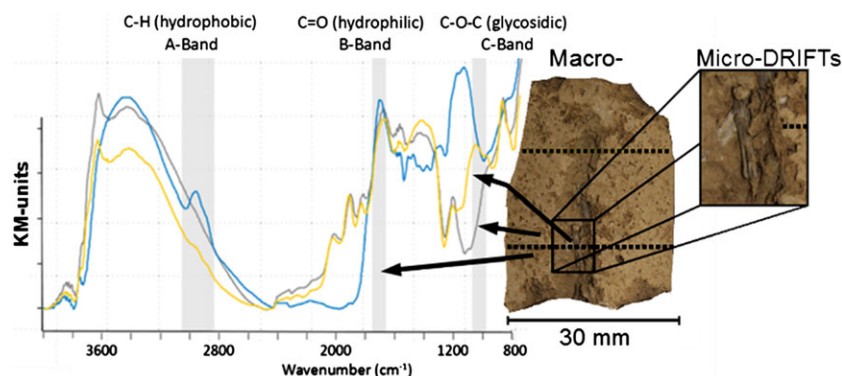


FIGURE 2 Exemplary spectra as derived from macro-diffuse reflectance infrared Fourier transform (macro-DRIFT) spectroscopy for particulate organic matter (i.e., straw residues; blue line), for soil matrix close to biopore surface (yellow line), and for the soil matrix more distant (grey line) to biopore surface. The right-hand side shows transects at biopore samples studied by macro- and micro-DRIFT spectroscopy. KM = Kubelka–Munk

the IR beam meets its designated position. Each spectrum obtained by micro-DRIFT was recorded by 32 co-added scans.

All DRIFT spectra, micro and macro, were recorded in a range of 400–4,000 cm^{-1} , with a resolution of 1 cm^{-1} (Fourier-transform infrared spectroscopy; Ellerbrock, Höhn & Rogasik, 1999), and corrected against ambient air as background by using a gold target (99% Infragold®, Labsphere, North Sutton, NH, USA) that was fixed onto the positioning table at the average sample surface elevation. Note, for macro-DRIFT, the diameter of the IR beam is about 1 mm (i.e., an area of 0.8 mm^2), whereas for micro-DRIFT, it is about 0.15 mm (i.e., an area of about 0.02 mm^2 ; Table 2). Therefore, the area of a single measurement point investigated by macro-DRIFT was about 40 times larger than that observed with micro-DRIFT.

All spectra were transformed into Kubelka–Munk (Kubelka, 1948, Kubelka & Munk, 1931) units using the software WIN-IR Pro 3.4 (Digilab, MA, USA), baseline corrected, smoothed (boxcar, Factor 15), and zapped for absorptions bands associated with carbon dioxide (wavenumber [WN] regions 2,386–2,297 cm^{-1}). Spectra from equidistant measurement points along transects were pooled, resulting in a much higher number of replicates for the surface of the biopore samples. The C–H band intensities of the DRIFT spectra—characteristic for hydrophobic alkyl groups in OM—were measured as a vertical distance (i.e., as height) from a local baseline plotted between tangential points in the spectral regions between WN 2,948–2,920 cm^{-1} and 2,864–2,849 cm^{-1} (Table 2). The C=O band intensities (i.e., characteristic for hydrophilic alkyl groups in OM) were measured as height from the total baseline of the spectra in the regions of WN 1,720–1,700 cm^{-1} and 1,625–1,600 cm^{-1} . The PWI (Leue et al.,

2013) used to characterize the OM was calculated as the ratio between the summed C–H signal intensities and the summed C=O signal intensities (C–H/C=O ratio; Ellerbrock et al., 2005). Larger PWI values indicate a smaller potential wettability of the OM.

2.4 | Soil water repellency (SWR)

Soil wettability was measured using a range of approaches (summarized in Table 3).

- I. An optical method provided the direct CA of intact soil samples by measuring the angle of a water droplet deposited on the soil surface by a syringe positioned above the sample. A high-speed camera (OCA20, DataPhysics, Filderstadt, Germany) with a frame rate of 250 s^{-1} was used to record droplet dynamics. SCA20 software (DataPhysics, Filderstadt, Germany) was used for analysing and calculating CAs. The soil samples' surface is not flat, which is especially true for the surface of biopores. To determine the CA correctly, we cut the samples into much smaller subsamples (≤ 5 mm thick $\times$ 10 $\times$ 10 mm), aiming to achieve plain sections. Air-dried subsamples were fixed to an aluminium plate. The left- and right-handed side CAs of each droplet were averaged. Measurements were replicated 3 times for each sample and each zone (bulk or biopore). Each droplet had a volume of around 8 μl . Under the assumption of a ball-shaped droplet, its extension in area is ≥ 4 mm^2 . Thus, this method showed the lowest spatial resolution applied on structured soils in this study. The used water was stored in vacuum to reduce the amount of dissolved gases in the liquid.

TABLE 2 Details about diffusive reflectance Fourier transform infrared spectroscopy (DRIFTS) for measurements at micro-scale and macro-scale. Macro-DRIFTS was performed with Bio-Rad FTS 135 (Bio-Rad, Munich, Germany), micro-DRIFTS with Cary 660 Fourier-transform infrared (IR) spectroscopy, connected to microscope model Cary 610 (microscope, Agilent Technologies, Santa Clara, CA, USA)

Parameter/method	Macro-DRIFTS	Micro-DRIFTS
Investigated area (mm^2)	0.79	0.018
Radii of IR beam (mm)	1	0.15
Max. distance to biopore (mm^{-1})	15	2.25
Smoothing factor (–)	15	
Wave number of functional groups (cm^{-1})		
C–H (asymmetric) “A” band	2,948–2,920	
C–H (symmetric) “A” band	2,864–2,849	
C=O “B” band	1,720–1,700	
C=O “B” band	1,625–1,600	

TABLE 3 Overview of the used methods and their spatial resolution

	Sessile drop method (goniometer)	Infiltration tests	Wilhelmy plate method	Diffuse reflectance infrared Fourier transform spectroscopy
Principle	Deposition of a water droplet, filmed by high-speed camera	Comparison of intrinsic infiltration rates of ethanol and water	Immersion of a defined plate into a liquid with known surface tension	Measures the diffusive reflection of defined wavenumbers
Results	Contact angles	Contact angles	Contact angles	Potential wettability index
Material	Intact soil cores	Intact soil cores	Sieved soil	Intact soil cores
Spatial resolution (mm ²)	>4.8	0.1	>26	0.8 (macro) or 0.018 (micro)

II. Wettability was also quantified from sorptivity tests at a hydraulic gradient of +0.02 m, as described in Hallett, Gordon, and Bengough (2003). Instead of a sponge filled tube, we used a smaller capillary (Minicaps, 5 μ l, Hirschmann Laborgeraete, Germany, ISO 7550) to gather infiltration rates of both water and ethanol successively with a spatial resolution of around 0.1 mm². Liquid uptake by the soil from a reservoir was recorded for 90 s every 1 s from a 0.1-mg balance. Sorptivity S (m s^{-0.5}) was calculated according to Equation 1 (Hallett et al., 2003):

$$S = \left(Q \cdot f \cdot (4 \cdot b \cdot r)^{-1} \right)^{0.5}, \quad (1)$$

where Q is the liquid flow rate (m³ s⁻¹), f is the fillable air-porosity (-) f (Leeds-Harrison, Youngs, & Uddin, 1994), b accounts for the wetting front and was assumed to be 0.55, and r is the diameter of the used capillary (m). The CA from sorptivity tests was calculated by

$$CA = \arccos \left(1.95 \cdot S_{\text{ethanol}} \cdot S_{\text{water}}^{-1} \right)^{-1}. \quad (2)$$

I. Measurements with the Wilhelmy plate method were performed on homogenized soils that were passed through either a 2-mm (Figure 6a) or 0.63- μ m (Figure 6b) sieve. The measurements were conducted as described in Goebel, Bachmann, and Woche (2008) and transformed to CA using calculations from Bachmann et al. (2003). We used a microtensiometer (Kruess, Hamburg, Germany) fitted with a Peltier element to maintain the temperature of the deionized water as wetting fluid at 20 °C. The Wilhelmy plate method provides both advancing and receding CAs, whereas most other methods only provide the advancing CA. Three replicates were measured for the bulk soil. As the method needs relatively large amounts of soil for a single measurement, we were not able to investigate biopore wall-borne soil. The CA (θ) was calculated according to Bachmann et al. (2003) from the force F (N) acting on the plate using Equation 3:

$$\cos \theta = F / (\rho \cdot \sigma_{LV})^{-1}, \quad (3)$$

with density ρ (Mg m⁻³) and surface tension of the wetting liquid σ_{LV} (N m⁻¹). The latter was determined using a microtensiometer and a platinum plate. The calculation according to Equation 3 is based on an assumed complete wettability of the platinum plate ($\cos \theta = 1$). For more details, see Holthusen, Haas, Peth, and Horn (2012).

2.5 | Statistical analyses

The statistical software R (R Development Core Team, 2017) was used for plotting and evaluating the results. CAs as derived from sessile drop method, from the Wilhelmy plate method, or from sorptivity tests, as well as water and ethanol sorptivity data, and the log-transformed PWI values were analysed with R software analogously.

The data evaluation started with the definition of an appropriate statistical mixed model (Laird & Ware, 1982; Verbeke & Molenberghs, 2000). Data were tested for normality and for homoscedasticity. These assumptions are based on a graphical residual analysis. The statistical model included (a) the history of the biopore (as shown in Section 2), (b) the sampling depths (0.45–0.55 m and 0.55–0.65 m), (c) the distance from the biopore surface, and (d) the pit (1–4), as fixed factors. The considered covariates (shown in Table 1 and their potential interaction effects) are based on a model selection, resulting in equations as follows:

$$PWI_{ij} = e^{a_{ij}} \cdot e^{b_{ij} \cdot \text{pH}_{ij}} \cdot e^{c_{ij} \cdot \text{dist}} \cdot e^{d_{ij}}, \quad (4)$$

with a , b , and c as fitting parameter, i and j define the histories of the biopores and the depths, and dist is the distance from the biopore surface (10⁻³ m). e is Euler's number.

The random effects were defined by the plots and suitable interaction effects with history, depth, and cylinder, assuming a split-plot design (pit + pit : history + pit : history : depth + pit : history : depth : cylinder + residual error). On the basis of this model, a pseudo R^2 was calculated (Nakagawa & Schielzeth, 2013), and an analysis of covariance was conducted (Cochran, 1957).

3 | RESULTS

For texture and soil organic carbon content, no depth-dependent trends were found (Table 1), whereas pH increased slightly with depth.

3.1 | OM composition and water repellency

Absorption bands of hydrophobic C–H groups (WN 2,948–2,920 cm⁻¹ and WN 2,864–2,849 cm⁻¹) were easily detectable in the spectrum of the biopores' surface and, especially, in the spectrum of particulate OM (Figure 2: blue line), but almost undetectable in the spectrum of the soil matrix (Figure 2: grey line). Furthermore, the DRIFT spectra of particulate OM were mostly not affected by so-called "bulk mode bands" (2,000–1,700 cm⁻¹, Ellerbrock, Leue,

& Gerke, 2016), caused by lattice vibrations in soil minerals (i.e., quartz) if those particles have diameters above 70 μm (Leue, Ellerbrock, Bänninger et al., 2010).

Because equidistant measurement points (i.e., with same distance from the surface of the biopore wall) were pooled at the macro-scale (Figure 3a), a much larger number of measurement points were considered for characterizing the biopore surface. The PWI values obtained from macro-DRIFT analysis showed highest means for the biopore surfaces (distance = 0 mm) and decreased with increasing distance from the biopore wall. The relatively high standard deviations for PWI (macro-DRIFT) indicate a large heterogeneity in quality and quantity of SOM distributed at the biopore surfaces. The PWI was found to scatter within 5- and 8-mm distance for the R-samples from Bt-2, within 1–3-mm distance for the EW samples from Bt-1 or within the 0–9-mm distance for the R-samples from Bt-1 and for both depths for the REW samples.

On the micro-scale (Figure 3b), the largest means of PWI were documented within some distance from the biopores surface: For the R-samples from Bt-1, two peaks in PWI were exhibited in 0.5- and 1.5-mm distance from the surface and smaller means for PWI from Bt-2. Pores colonized by earthworm showed larger values within 0 to 1.5 mm (Bt-1) or 0 to 0.6 mm (Bt-2) in distance from the biopores' surface. Values for REW from Bt-1 showed the most pronounced scattering with its maximum in 0.15-mm distance from the biopores' surface or an almost absence of larger PWI values within 1.5 mm in distance from the biopores surface and a peak in 1.75 mm (Bt-2). On the micro-scale, REW showed the smallest depth-dependent means for the PWI of all biopore histories.

The statistical analysis of covariance (see Section 2.5) showed significant interactions for the logarithmic PWI between the history, the distance, and the depth on the micro-scale. At the macro-scale, the influence of the pH value improved the statistical model, additionally. Table 4 shows the regression coefficients for macro-scale that can be used with Equation 4. Because the statistical analysis for results of the sessile drop method showed that the sampling zone (matrix or

TABLE 4 Correlation coefficients (a , b , and c) as derived from analysis of covariance for results derived from macro-diffuse reflectance infrared Fourier transform spectroscopy. Coefficients according to Equation 4 for biopores colonized and/or created by chicory (*Cichorium intybus* L.) root (R), colonized and/or created by *Lumbricus terrestris* (EW), and for biopores, colonized and/or created by a plant root followed by colonization of *L. terrestris* (REW) for 0.45–0.55 m (T1) and 0.55–0.65 m (T2). Degree of freedom (dF) for a and c dF = 6 or dF = 546 for b . $R^2 = 0.38$

	R		EW		REW	
	T1	T2	T1	T2	T1	T2
a	3.284					
b	−28.441	−28.567	−28.129	−28.795	−28.645	−28.530
c	−0.114	−0.142	−0.196	−0.115	−0.102	−0.119

biopore) was the only significant influence on CAs, we pooled the biopores' histories (Figure 4). This resulted in 8 points for each zone (biopore and bulk), each representing a single depth of each pit. In Figure 4, the PWI is plotted against the CA as derived from the sessile drop method (Figure 4a) or sorptivity tests (Figure 4b). The sessile drop method delivered promising results with highly significant differences between biopores and bulk soil. Here, the means of the bulk soil CAs are close to 30° and could be determined with good repeatability as indicated by the small standard deviations. The biopores' heterogeneity is reflected by means of CAs between 55° and 85° .

CA derived from sorptivity tests (Figure 4b) differed from those of the sessile drop, and no clear statistical influence of the type of surface (matrix or biopore) was found. There were only weak ($0.05 \leq p \leq 0.1$) significant interactions between the CA and matric potential, the measurement zone and the pit the soil cores were from. The CA values derived from sorptivity tests ranging from $46.9^\circ \pm 29.1^\circ$ to $70.3^\circ \pm 6.4^\circ$ (Table 5) of matrix surfaces are not significantly different from those of biopore surfaces (ranging from $48.8^\circ \pm 17.1^\circ$ to $75.2^\circ \pm 4.6^\circ$). The same is found for the water sorptivity (Figure 5), which ranged from 0.002 to 0.07 $\text{mm s}^{-0.5}$. However, general trends could be observed: With decreasing water content (decreasing matric potential), scattering and

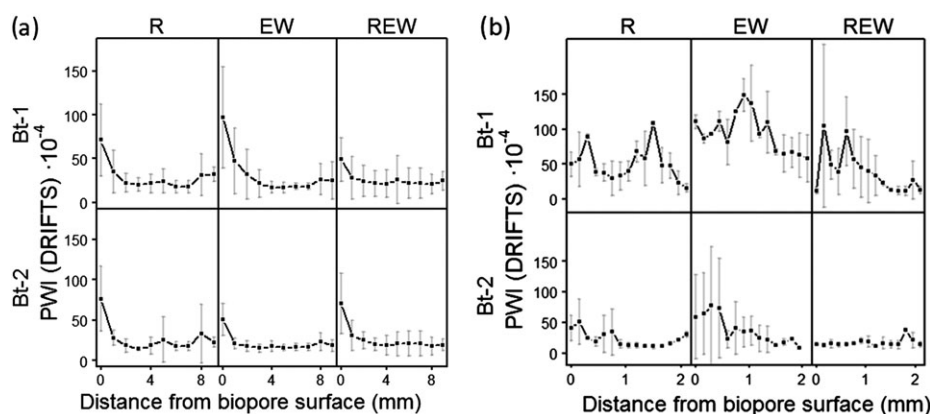
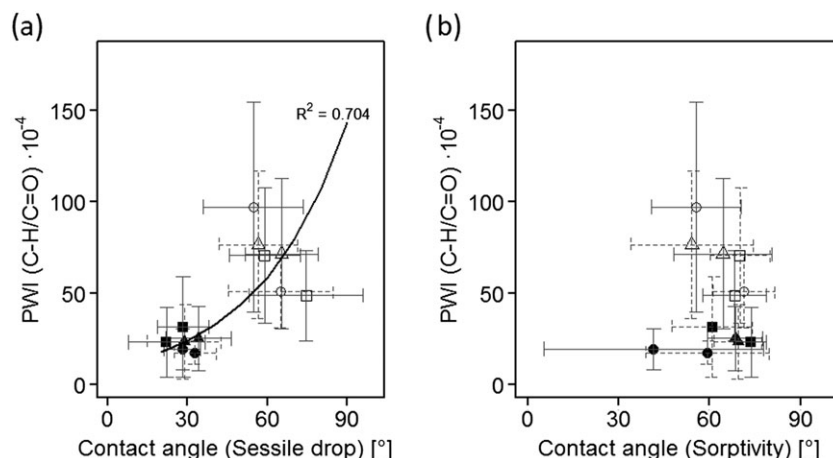


FIGURE 3 Means and standard deviation for potential wettability index (PWI), derived from (a) macro- and (b) micro-diffuse reflectance infrared Fourier transform (DRIFT) spectra of biopores colonized and/or created by *Lumbricus terrestris* (EW), colonized and/or created by chicory (*Cichorium intybus* L.) roots (R), and biopores colonized and/or created by a plant root followed by colonization of *L. terrestris* (REW). The upper row shows data for Bt-1 (0.45–0.55 m) whereas the lower row those for Bt-2 (0.55–0.65 m). PWI is defined as the ratio of the intensities of summed C–H signals to summed C=O signals. Results of measurements at the biopores' surface were pooled (=0-mm distance from biopore surface). The same was done for equidistant measurement points. For (a) $n_{R,Bt-1} = 29,8,8,8,8,8,7,7,5,3$; $n_{EW,Bt-1} = 37,8,8,8,8,8,8,6,4$; $n_{REW,Bt-1} = 34,8,8,8,8,8,8,5,3$; $n_{R,Bt-2} = 28,8,8,8,8,8,7,7,4,3$; $n_{EW,Bt-2} = 33,8,8,8,8,8,8,5,6$; $n_{REW,Bt-2} = 35,8,8,8,8,8,8,4,8$. For (b) $n = 4$

FIGURE 4 Relation between potential wettability index (PWI) derived from macro-diffuse reflectance infrared Fourier transform spectroscopy (Y axis) and contact angles derived from (a) optical measurements (sessile drop) on bulk soil (distance from biopore surface ≥ 5 mm, filled symbols) and biopore surface (blank symbols) or (b) sorptivity tests. The biopores' histories were presented by differing symbols (colonized and/or created by chicory (*Cichorium intybus* L.) root = triangle, colonized and/or created by *Lumbricus terrestris* = dots, and colonized and/or created by a plant root followed by colonization of *L. terrestris* = squares). Shown are arithmetic means with one standard deviation (solid line for Bt-1 and dashed line for Bt-2). Results of measurements at the biopores' surface were pooled (≈ 0 -mm distance from biopore surface). Shown sorptivity data are for $\Psi_m = -30$ kPa, determined at a hydraulic gradient of +2 cm with an infiltrometer device, 0.22 mm in diameter. $n_{\text{sorptivity}} = 3$



water sorptivity increases, whereas differences in median between matrix and biopore surfaces disappear. Note, pores colonized by earthworms (i.e., EW; $\Psi_m = -1$ kPa, and REW; $\Psi_m = -3$ kPa) show a good repeatability for sorptivity data.

The CA values as derived from the Wilhelmy plate method ranged from 38.4° to 66.6° , if the soil is sieved to ≤ 2 mm (Figure 6a) or from 30.4° to 52.9° , if the soil is sieved to ≤ 0.63 mm (Figure 6b). Although the latter one showed a good repeatability indicated by lower standard deviations, the values were not different for biopore as compared with matrix surface samples.

4 | DISCUSSION

The presented water sorptivity rates are more than 10 times lower than those shown by Hallett et al. (2003) who investigated soil from both the rhizosphere and bulk soil at different water contents. However, in contrast to Hallett et al., 2003, who used an infiltrometer device with 0.4 mm in diameter, the infiltrometer tip used here shows a diameter of about 0.22 mm. Note, as stated by Hallett et al. (2003), the sorptivity rates should decrease with decreasing diameter of the infiltrometer tip that is in accordance to the 10 times lower sorptivity

rates found here. The larger repeatability of the water sorptivity rates (Figure 5) for the surface samples from EW and REW could be explained by homogenization caused by the earthworms' activity. However, differences in water transport were not statistically significant. These differences could be caused by two processes: (a) the colonizers' influence on aggregation and (b) exudates and secondary metabolites (Hallett et al., 2003) because they may coat soil particles

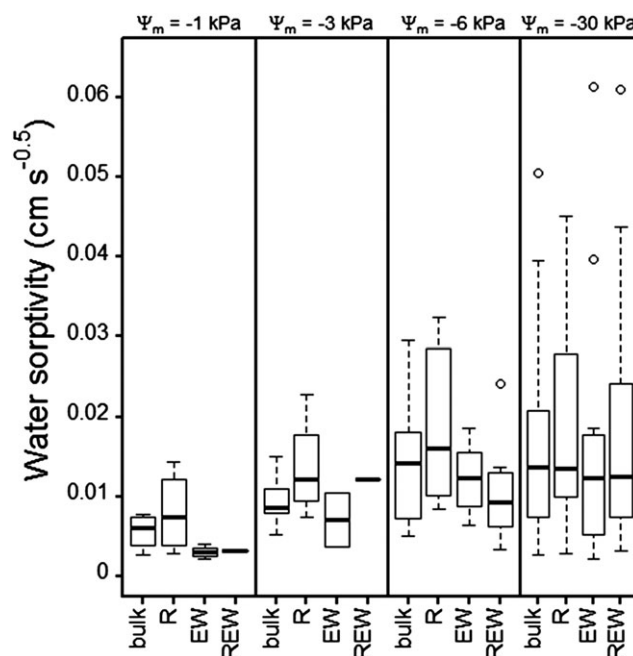


FIGURE 5 Influence of matric potential and zone (bulk soil or biopore with defined history (colonized and/or created by *Lumbricus terrestris* [EW], colonized and/or created by chicory (*Cichorium intybus* L.) root [R], and for biopores, colonized and/or created by a plant root followed by colonization of *L. terrestris* [REW]) on water sorptivity, determined at a hydraulic gradient of +2 cm with an infiltrometer device, 0.22 mm in diameter. $n_{\text{sorptivity}} = 3$

TABLE 5 Means and standard deviation for contact angles derived from sorptivity tests for biopores colonized and/or created by chicory (*Cichorium intybus* L.) root (R), colonized and/or created by *Lumbricus terrestris* (EW), and for biopores, colonized and/or created by a plant root followed by colonization of *L. terrestris* (REW)

	Contact angle [°]			
	-1 kPa	-3 kPa	-6 kPa	-30 kPa
R	48.8 ± 17.1	65.7 ± 14.3	69.4 ± 8.6	56.0 ± 27.7
EW	67.0 ± 16.3	56.3 ± 16.7	75.2 ± 4.6	66.8 ± 13.6
REW	70.0 ± 4.1	72.3 ± 7.7	65.3 ± 12.9	73.0 ± 4.8
Soil matrix	46.9 ± 29.1	65.3 ± 14.4	66.0 ± 10.9	70.3 ± 6.4

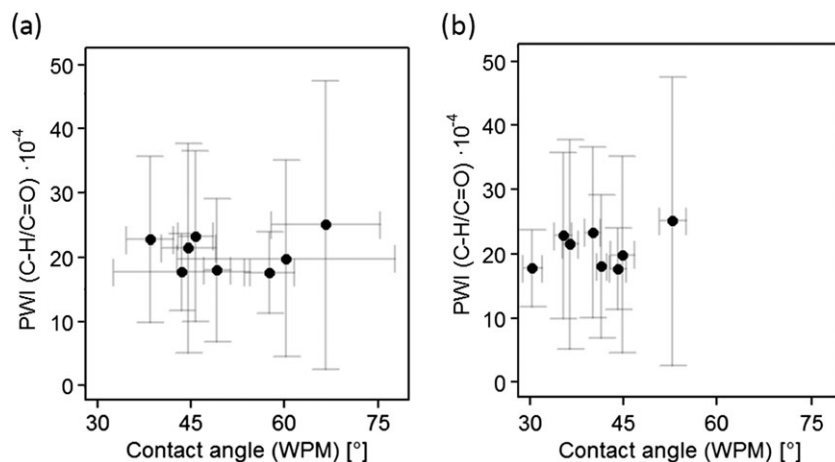


FIGURE 6 Relation between potential wettability indexes (PWI) as derived from macro- diffuse reflectance infrared Fourier transform spectroscopy and contact angles derived from Wilhelmy plate method for soil sieved to ≤ 2 mm (a) or ≤ 0.63 mm (b). Means with standard deviation for each depth and plot of the bulk soil (distance to biopore surface ≥ 5 mm). $n_{\text{WPM}} = 3$

and induce slightly higher water repellency in the outermost layer of biopore samples (Czarnes, Hallett, Bengough, & Young, 2000). The influence of such outermost layers could be shown by comparing samples ≤ 2 mm (Figure 6a) with those ≤ 0.63 mm (Figure 6b). By sieving the soil to 0.63 mm, aggregates >0.63 mm were destroyed and therefore with their organic coatings, leading to a more homogenous soil. This homogenization resulted in lower means and standard deviations (effect of dilution of hydrophobic substances). It could be assumed that the hydrophobicity of the aggregate coatings is mostly caused by two processes: (a) the accumulation of amphiphilic organic substances that are not able to diffuse into the aggregates' interior due to their size and (b) due to differences in the spatial orientation of SOM functional groups caused by the dual porosities of soils (aggregates' in and exterior pore space): If a water-saturated soil dries, its drainage starts with its largest pores. In this case, the hydrophobic functional groups of SOM are orientated towards air-filled pores, whereas the hydrophilic functional groups are orientated towards the soil water (Bachmann et al., 2003). The largest pores are mostly biopores, whereas coarse pores are found at the interaggregates' pore space, which is located next to the aggregates' exterior. The smallest pores were found at the aggregates' interior. Because some of these pores never dried, this change in the spatial orientation of the hydrophilic groups in SOM towards the mineral particles is not forced as it is for the larger pores, resulting in lower CA. However, the results for the homogenous soil fit well to those gathered from the sessile drop method.

As in Leue, Ellerbrock, and Gerke (2010), Leue et al. (2013), and Fér, Leue, Kodešová, Gerke, and Ellerbrock (2016), we found larger PWI values for the surfaces of root channels and earthworm burrows, as compared with the matrix (i.e., surrounding soil). Such higher PWI values are also related to a lowered wettability of these regions, described by increased CAs (Leue, Gerke, & Godow, 2015) resulting in decreased infiltration rates (Hallett et al., 2003). The lower wettability of the biopore surfaces may be caused by the accumulation of organic components (such as plant waxes or terpenes) that originate from earthworms or roots by release and/or illuvial processes and perfectly follows the enlarged transportability of these organic compounds, caused by increased diffusion coefficients in biopore walls compared with soil matrices as found by Koebernick et al. (2017) and Haas and Horn (unpublished).

The PWI values determined from macro-DRIFT spectra of earthworm (EW) burrow walls are similar to those of the root channels (R; Figure 3a) but larger than those of root channel surfaces recolonized by *L. terrestris* (REW), whereas PWI values determined from micro-DRIFT spectra indicate a strong decrease in PWI for the REW from Bt-2 as compared with R and EW samples (Figure 3b). Two processes may be responsible: an enhanced mechanical redistribution of organic particles caused by a total loss of soil aggregation, which may be the result from shearing forces exerted by earthworm movements through thickening and following sliding through the cracking soil volume at high water contents. Because of homogenization caused by the earthworm's activity, regions with higher means in PWI (surface of biopores and/or exterior of aggregates) are mixed with regions of low PWI (soil matrix and/or interior of aggregates) resulting for REW samples in PWI values lower than for EW and R samples. We expect the homogenization potential of a moving earthworm to be larger at biopores that had been colonized by plant roots prior to earthworm colonization, because plant roots alter the soil surface chemically and physically (Ruiz, Or, & Schymanski, 2015). Exudates increase the water content at given matric potential; nutrient uptake affects the surface charge and pH apart from a physical pore-size shift and altered pore geometry (Whalley et al., 2005). Consequently, the decay of organic particles should be enhanced in biopores colonized by both EW and R, because such biopores show pH values more favourable for micro-organisms than biopores colonized by R or EW alone (Whalley et al., 2005). Biopores colonized only by roots will be more acidic due to CO_2 and H^+ release, whereas those colonized solely by EW will result in a more alkaline soil due to calcite release from specialized oesophageal glands (Lankester, 1865). According to DRIFT data, the zone of the biopore walls influenced either by EW, R, or REW is limited to a thickness of about 0–2 mm that is in accordance with Koebernick et al. (2017), Uteau et al. (2015), and Leue, Ellerbrock, Bänninger, et al. (2010) and Leue, Ellerbrock, and Gerke (2010). For biopores created by EW and REW, the PWI values became relatively constant (i.e., small standard variations) for zones that are in distances above 2 mm from the biopore wall (Figure 3), whereas the PWI for <2 mm in distance from the pore wall surface indicated the formation of a kind of lining. Such trend was especially detectable with micro-DRIFT technique (micro-scale, Figure 3b) for EW biopores. However, for the R-biopores at distances above 2 mm, strong variations in PWI values were found at both, macro

and micro, scales. Such stronger variation in PWI for the R-biopores may be explained by the effects of lateral roots growing into the soil matrix surrounding the root channel. Differences in the width of the earthworm linings (0 to 1.5 mm for Bt-1 and 0 to 0.6 mm for Bt-2) can be explained by a higher energy needed for the mechanical work to extent the cavity with increasing soil depths (as described by Mohr–Coulomb theory, Horn et al., 1994; Ruiz et al., 2015).

The results of sorptivity test and sessile drop procedure (Figure 4) confirmed the PWI determined from DRIFT spectra, indicating a reduced wettability for biopore walls. This is in accordance with findings of Hallett et al. (2003) using sorptivity tests and with those of Leue et al. (2015) using sorptivity tests and sessile drop method.

Differences between the results of the three different CA measurement methods and the PWI are caused by (a) the different spatial resolutions and (b) methodological limitations of the used techniques, as well as by (c) differences in sample preparation (intact, and sieved samples). Using the identical technique, the obtained results are albeit different because of the spatial distribution of organic components such as SOM at surfaces of intact soil samples. It is distributed mostly heterogeneous (i.e., patchy layers; Kaiser & Guggenberger, 2000) and has especially with a highly sensitive method, a large impact on the measured parameters (e.g., PWI values of micro- vs. macro-DRIFT). If different methods are used and the results compared, also different results will be obtained for sorptivity tests because these procedures are mostly limited to samples that show CA smaller than 90° because the samples need to be wettable to a certain extent (Bachmann et al., 2003). In contrast, both the sessile drop and the Wilhelmy plate method allow to analyse samples with CA from 0° to 180°. Cosentino, Hallett, Michel, and Chenu (2010) compared different methods for assessing the wettability of soils (viz., sorptivity tests, capillary rise method (CRM), and water droplet penetration time test) and stated that these methods measure different related soil properties. In soils with small levels of water repellency, water droplet penetration time test determines the rate of wetting by water, whereas the R index (calculated from sorptivity tests) and CRM measure hydrophobicity. Furthermore, the sorptivity test uses two wetting liquids, with ethanol potentially mobilizing organic compounds, changing pore surface roughness and causing differences in swelling at the wetting front compared with water. Furthermore, as Leue, Ellerbrock, and Gerke (2010) stated, the CA is characterized by the first contact of a droplet at a sample surface; thus, the OM properties of the outermost surface molecules are of essential importance (Bachmann et al., 2003). Fungal hyphae are, for example, known for their hydrophobicity. In this study, the maximum CA of about 132° was measured (sessile drop method) directly at a fungal mycelium. Such data show extreme hydrophobic behaviour, whereas fungal hyphae can be also partly hydrophilic as well that was observed by goniometer measurements in this study. Such hyphae were roped into a water droplet, followed by a release of spores. The processes result in smaller CA values. Additionally, inconsistencies between CA and PWI data are probably caused by the penetration depth of the IR radiation of about 6 to 8 µm that strongly exceeds the thickness of the outermost molecular OM layer (Diehl, 2013). The results show that the extent of water repellency of a biopore wall finally depends on their colonization history (root and/or earthworm) and on the soil depths and is much more pronounced

in biopore walls than it is in the soil matrix. Therewith, the capillary rise or the water content at a given matric potential is lowered (described by Jurin's law) and so is the flow of water through the biopore wall at a given matric potential. However, small-scaled (micro) DRIFT measurements showed that even if macro-DRIFT measurements indicate pronounced water repellency, biopore walls can have a certain surface percentage with less pronounced water repellency. These regions act as preferential flow paths from the biopore wall into the soil matrix. Finally, we also need to consider the time dependency or reversibility of the water repellency that affects the water flow during an event of precipitation. However, until now, these dynamic processes are neither completely understood nor have interactions between the composition of mineral surface layers and time-dependent alterations of water repellence been quantified. Further research should therefore focus on the dynamics of OM quality and water repellency in biopore walls and their surrounding soil matrixes. The repeated analysis of defined loci of artificial pores that could be colonized by an earthworm and/or root at defined time points during the recolonization would eliminate the influence of soil heterogeneity.

5 | CONCLUSION

1. The CA, the PWI, and with that the chemical composition of biopore walls differed from the soil matrix. Furthermore, the spatial extent of the alteration of the soil depends on the biopores' colonizer (root, earthworm, or both) and on further soil factors such as soil depth.
2. Results showed that SWR is heterogeneously distributed at transects across biopores. Thus, water flow is expected to occur preferentially through regions with a less pronounced SWR.
3. The linkage of PWI values to CAs gathered from other methods (especially from the sessile drop method/optical measurements) delivered promising results despite methodologically given differences in spatial resolution.
4. The DRIFT spectroscopy is an easy, fast, and reliable method that allows feedbacks on the chemical composition of the organic coatings of surfaces. Its potentially high spatial resolution and its broad field of application make the DRIFT spectroscopy a highly appropriate approach in small-scaled soil analysis. However, the PWI, a dimensionless parameter, indicates only a potential for water driven transport but did not reflect real transport functions.
5. Further research is needed to derive transfer functions that link PWI values with wettability properties such as CA.
6. Earthworm activity was found to reverse the reduced wettability (in terms of high means in PWI) of biopore walls, resulting in lower CA and therewith higher infiltration rates into the surrounding soil that in turn leads to a larger amount of water available for plant production. Consequently, we suggest an increase of both the number of deep rooting plants in crop rotation and the number of individuals of earthworms to optimize water transport or even capillary rise through agricultural soils with respect to sustainable agricultural productivity. This can be justified using Good Agricultural Practices.

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